

MOVABLE ANTENNA ENABLED DUAL-SCALE BEAMS FOR RESILIENT COMMUNICATION, SENSING, AND SECURITY

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ABSTRACT

Upcoming sixth-generation (6G) wireless networks are expected to provide ultra-high data rates, precise sensing capabilities and strong security. Movable antenna (MA) technology emerges as a promising solution by dynamically reconfiguring antenna positions in real-time, thereby exploiting spatial degrees of freedom and aligning with the communication-sensing-security framework integral to 6G. The core concept of this study focuses on MA-enabled dual-scale beams, a novel framework that synergistically combines wide and narrow beams to enable adaptation across various scenarios. The paper begins by presenting the principles of MA-enabled dual-scale beams and their application scenarios. We proceed to discuss resilient communications and the design of dual-scale beams to enhance communication resilience. By integrating wide and narrow beams, we explore a two-phase sensing strategy to improve sensing agility and precision. Through wide beam isolation and narrow beam encryption, dual-beams can strengthen the physical layer security for massive users. Finally, we explore future research directions to advance the implementation of dual-scale beam technology.

I. INTRODUCTION

The deployment of fifth-generation (5G) wireless networks has significantly enhanced user experience. Nonetheless, as communication technologies advance rapidly, new applications like autonomous driving, robot navigation, and virtual reality emerge demanding ultra-high data rates, precise sensing, and secure connectivity [1], [2]. These demands are driving the evolution from 5G to sixth-generation (6G) wireless communication systems. As a result, there is an increasing interest in the novel integration of communication, sensing, and security in future wireless networks.

With advances in the multiple-input multiple-output (MIMO) technologies, the capacity of today's

wireless communication systems has been dramatically increased by exploiting the new degrees of freedom (DoFs) in the spatial domain. By leveraging the multipath components, MIMO or massive MIMO technologies can improve the spectral efficiency beyond that of single-antenna systems [3], [4]. Nevertheless, conventional MIMO or massive MIMO systems usually adopt fixed antennas (FAs) with spacing no smaller than a half-wavelength at the transceivers. Fixed arrays cannot physically reconfigure their geometry to respond to real-time changes in user distribution, target mobility, or signal propagation dynamics. Even antenna selection schemes, which switch between predefined static positions, fail to unlock the full potential of spatial resource utilization. Thus, such fixed and discrete deployment of antennas limits the diversity and spatial multiplexing performance of MIMO systems because the channel variation in the continuous spatial field is not fully utilized.

Movable antenna (MA) technology provides an innovative approach to address the limitations of traditional FA architectures in 6G networks [5], [6]. Unlike conventional FAs, MAs can be repositioned within a spatial area to fully utilize spatial resources, offering several benefits [7]. Firstly, expanding the movement area of antennas effectively increases the array aperture, thereby enhancing angle estimation resolution [8]. Secondly, optimizing the MA geometry reduces the correlation among steering vectors in various directions, reducing estimation ambiguity and interference. Thirdly, MAs can be adjusted in real-time to accommodate changing environmental conditions and sensing requirements [9]. Additionally, adaptable antennas facilitate enhanced beamforming, developing anti-eavesdropping capabilities and reinforcing physical layer security [10].

MA-based communication, sensing and security systems exhibit significant benefits and the potential for performance improvements. Most existing studies primarily focus on static environments. However, MAs deliver greater advantages in dynamic environments. In this paper, we propose an innovative dual-

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scale beam framework integrating wide and narrow beams for adaptability across various scenarios. We present the MA-enabled dual-scale beams and their application in advancing communication, sensing, and security. Subsequently, we explore resilient communication strategies and the design of dual-scale beams adaptable to changing conditions. To achieve precise sensing, a two-phase strategy is proposed, involving wide beam scanning and narrow beam localization. For physical layer security, wide beams facilitate user isolation, while narrow beams secure data transmission. Finally, conclusions and future research directions are discussed.

II. DUAL-SCALE BEAMS AND ITS APPLICATIONS

In this section, we briefly introduce the beamforming technologies. Then, we analyze the dual-scale beams by using movable antennas and present the applications of dual-scale beams for wireless communication, sensing, and security.

A. BEAMFORMING TECHNOLOGIES

A single antenna element radiates energy omnidirectionally, while multiple elements are combined into an array to unlock the powerful capability of beamforming, which generates focused beams of electromagnetic energy to the target user. Beamforming relies on the core principle of wave superposition. When multiple antenna elements radiate simultaneously, their individual electromagnetic fields combine in space. The radiation (or reception) pattern of the antenna group is considerably different from that of a single antenna. This is due to the constructive and destructive interference properties of radio waves. Depending on the relative phase and amplitude of the signals fed to each element, this superposition can be manipulated to create constructive interference in desired directions and destructive interference in others.

The antenna array response of multiple identical elements is usually characterized in terms of the array factor (AF), which is the complex-valued far-field radiation pattern obtained for an array of isotropic radiators, measuring how much a specific characteristic changes because of the grouping. The overall beam pattern of an array is the product of the radiation pattern of a single element and the array factor. The AF is purely determined by the geometry of the array (positions of elements) and the complex weights (amplitude and phase) applied to each element. The most common method for beamforming with FAs is precoding, which is a signal processing method widely used for optimizing signal strength and directionality. By adjusting the signal's phase and amplitude for each antenna element, specific direction, rather than broadcasting signals equally in all directions, helps enable efficient spectrum use, reduce interference, and enhance network performance and reliability.

Although beamforming can generate focused beams to the targeted user, some beam characteristics cannot be changed only by beamforming, such as the beamwidth. The beamwidth is defined as the sharpness (width) of the main beam, which is inversely proportional to the array's electrical size. Larger arrays (more elements or larger overall dimensions) produce narrower beams and higher directivity. That is the reason why massive antennas can provide higher spatial diversity and improve the communication performance. While powerful, beamforming

with FAs relies on fixed antenna geometries, limiting adaptability in complex real-world scenarios.

B. MA-ENABLED DUAL-SCALE BEAMS

To satisfy various requirements of different scenarios, we propose a dual-scale beam structure. The dual-scale beam refers to narrow beams and wide beams. Narrow beams focus energy in specific directions, achieving high-precision signal transmission. This results in improved signal quality and enhanced spectral efficiency, making them ideal for high-capacity data transmission, high-precision sensing, and secure communication links. Wide beams, on the other hand, provide broad coverage, enabling rough positioning, maintaining basic connectivity in dynamic environments with high mobility. Narrow beams optimize performance for common users, while wide beams handle some special situations. This dual-scale approach enhances overall system capacity, reliability, and user experience, meeting the diverse requirements of various communication scenarios and applications.

For the dual-scale beams, the MA is the key component, which is an antenna system where the radiating antennas can be dynamically repositioned within a defined spatial domain [11]. By installing the MA on a 3D mechanical slide, the central processing unit controls the antenna position by driving step motors. The motors can cooperatively perform the corresponding steps to relocate the antenna to a target position [12]. Antenna positions do not need to be adjusted for each time slot. Instead, MAs can be designed and adjusted over a longer period based on scenarios and communication conditions, with beamforming technology used for further beam adjustments. As mentioned before, some beam characteristics such as beamwidth, depend on antenna geometries. The physical mobility of MAs directly changes the antenna's spatial geometry, enabling real-time optimization of beamwidth. Unlike beamforming altering phase shifts electronically, MAs achieve beam control by changing the physical path lengths between elements. This shift from electronic phase adjustment to spatial position optimization enhances the system's spatial DoF, allowing it to adjust channel conditions, thereby potentially improving signal intensity and reducing attenuation. Table 1 summarizes the differentiators of fix antennas and movable antennas.

Fig. 1 presents the array factor of different antenna geometries, including uniform antenna spacing and non-uniform antenna spacing of 4 antennas. For uniform antenna spacing, the small spacing antenna array can generate a wide beam, while the large spacing antenna array can generate a narrow beam. For the cases of (b), (c), and (d), the antenna arrays have the same antenna aperture. The main lobe patterns of these three antenna arrays are similar. However, different antenna positions can generate different side lobe patterns. MAs, combined with precoding, can generate various beams to meet diverse requirements in different scenarios of communication.

C. DUAL-SCALE BEAMS FOR COMMUNICATION, SENSING, AND SECURITY

Conventionally, narrow beams concentrate energy toward specific directions, which can strengthen the received signals and reduce the mutual interference,

providing significant benefits in communication, sensing, and security. Thus, for instance, massive MIMO, leveraging large-scale antennas to generate narrow beams, can enhance system performance. However, in situations like high user mobility, narrow beams cannot point to the accurate positions, leading to performance loss, while wide beams may preserve system stability. Next, we will explore dual-scale beam applications for communication, sensing, and security, as illustrated in Fig. 2.

Enhanced Communication: The 6G wireless networks are expected to provide ultra-high transmission rates and support a massive number of various connected applications. Conventional narrow beams, which can be precisely directed at specific users, improve signal strength while reducing interference, suitable in simpler scenarios. However, they struggle in high-mobility scenarios where maintaining connections with fast-moving users is challenging. Dual-scale beams using MA optimize antenna and beam configurations. By generating a wide, robust beam, the base station maintains connections with high-speed users. Consequently, dual-scale beams are used in an adaptive manner, switching narrow or wide beams to align with channel conditions and user needs, thereby ensuring robust, high-capacity links in dynamic settings that outperform FA systems in terms of coverage, capacity, and energy efficiency.

Advanced Sensing: The 6G network is expected to feature sensing capabilities that cater to emerging applications in need of high-precision positioning and mapping. Narrow, focused beams naturally enhance directional accuracy and spatial resolution, facilitating precise detection and localization of objects. However, without prior knowledge of object positions, locating targets with narrow beams can be both time-intensive and inefficient. Furthermore, in environments with multiple targets, narrow beams can suffer from spatial ambiguity due to periodic phase differences. Dual-scale beams are sequentially employed in sensing scenarios, transitioning from broad situational awareness to precise localization, to offer flexible sensing capabilities.

Strong Security: The advancement of communication networks has elevated security to a crucial concern. Beyond traditional security challenges, 6G is poised to encounter new threats, necessitating the consideration of strong security in its design. Physical layer security utilizes the wireless channel characteristics to ensure confidential data transmission, where narrow beams can exploit the superior quality of the legitimate user's channel over that of an eavesdropper or generate symmetric encryption keys from reciprocal channel observations, showing significant promise for improving security. However, with a large number of users, designing narrow beams is challenging due to the difficulty in preventing interception by curious users. Dual-scale beams provide a collaborative defense strategy. By simultaneously employing dual-scale beams, wide beams divide users into separate regions for spatial isolation, while Within each isolated region, narrow beams provide directional encryption. These dual-scale beams are formed via two-stage precoding. This dual-scale approach—combining spatial separation via wide beams with directional signal focusing through narrow beams—forms a strong security system guarding against interception and jamming attacks.

Feature	Fixed Antennas	Movable Antennas (MAs)
Beam Control	Electronic	Physical and Electronic
Beam Scaling	Limited by antenna aperture	Flexible
Degrees of Freedom	Limited by the number of antennas	Dramatically high
Coverage Range	Constrained by array geometry	Extendable via physical movement
Beam Adaptability	Fixed beam pattern	Dynamic 3D beam
Channel Change	Cannot change channel	Construct Channel
System Performance	Moderate	High
Hardware Complexity	Low	High

TABLE I. Key differentiators between fixed antennas and movable antennas.

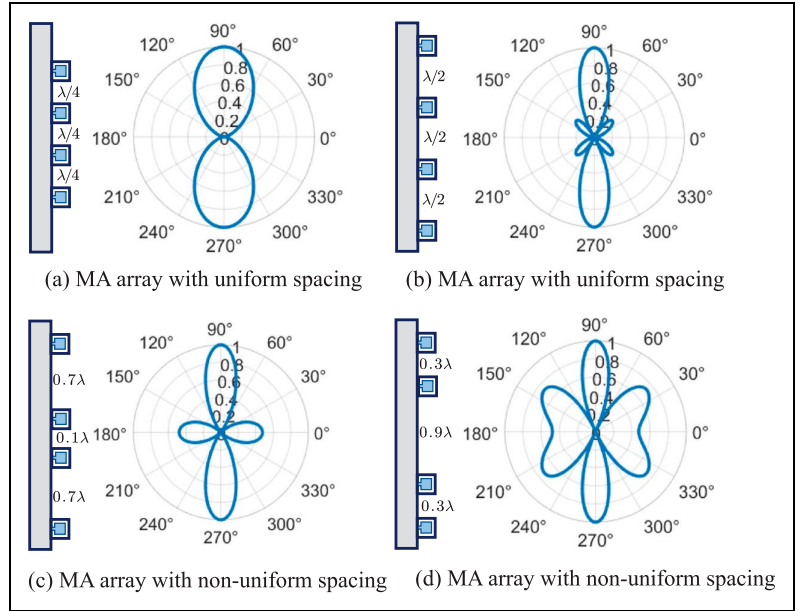


FIG. 1. Array factor of different geometries of movable antennas.

III. DUAL-SCALE BEAM ENHANCED RESILIENT COMMUNICATION

In this section, we mainly analyze the challenges for wireless communications under some special scenarios and introduce the resilient communications. Then, we discuss the dual-scale beams system design to enhance the resilient communications.

A. RESILIENT COMMUNICATION

The advancement of communication technology has enabled the existing mobile communication to satisfy most daily communication needs, such as video streaming, voice calls, and online games. However, performance degrades sharply in complex, dynamic environments characterized by high-speed mobility or signal interruptions. In these situations, rapid channel changes, significant user interference, and sudden deep signal fading result in unreliable connectivity, decreased data rates, increased latency, and service disruptions. Therefore, robust communication necessitates a shift towards resilient wireless systems capable of autonomously adapting to, maintaining, and recovering performance despite severe impairments.

Resilient communication refers to the capability of a wireless system to maintain stable and reliable data

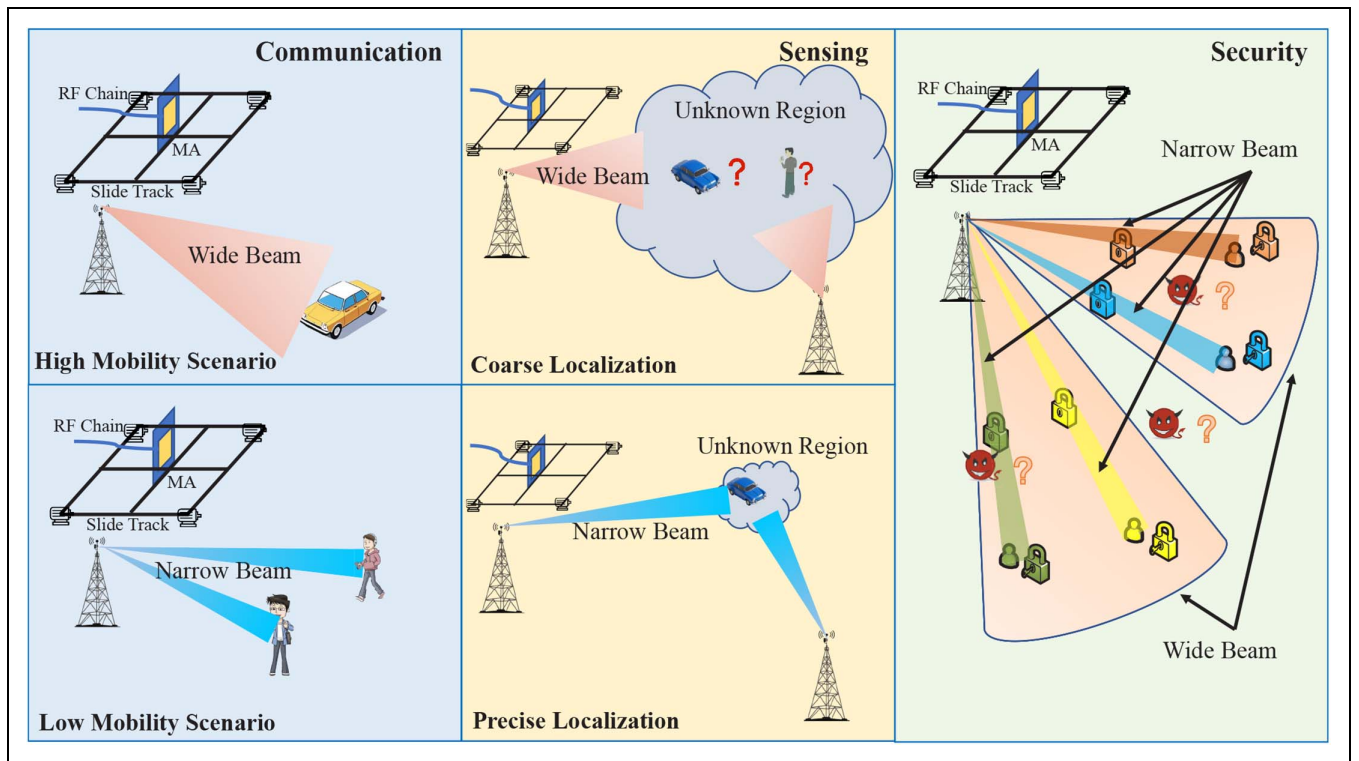


FIG. 2. Application scenarios of dual-scale beams.

transmission even when channel conditions deteriorate significantly (e.g., due to fading, interference, blockages, or high mobility). Crucially, it exhibits two key characteristics:

- 1) Graceful Degradation: When a wireless system encounters challenging conditions such as fading, interference, obstructions, or high mobility, its performance metrics should not fail suddenly. Instead, they should exhibit a controlled and minor degradation. For instance, the data rate may slightly decrease and packet success rate could drop marginally. By ensuring graceful performance decline, users can maintain communication, preventing total failures and ensuring a more consistent user experience.
- 2) Rapid Recovery: As adverse conditions alleviate, it must promptly regain near-optimal performance to maintain communication efficiency. It is essential for the system to avoid prolonged suboptimal states post-challenge, adapt quickly, and restore high-quality communication, thereby ensuring consistent effectiveness and reliability in fluctuating environments.

Resilient communication is attainable via dynamic, multi-layered adaptation which includes environmental awareness and flexible configuration of spatial, temporal, and frequency resources. Initially, real-time sensing monitors the environment, identifying impairments like fading, interference, and mobility. Then, spatial, temporal, and frequency resources are modified to ensure communication stability. Notably, by employing MAs to physically adjust antenna positioning and beamforming, dual-scale beams can dynamically steer spatial signals, ensuring uninterrupted, reliable service within changing environments.

B. DUAL-SCALE COMMUNICATION BEAM DESIGN

The dual-scale communication beam utilizes MA technology to dynamically adjust beam characteristics,

enabling adaptive signal coverage across varying communication settings. Wide beams trade off peak gain for broader angular coverage to preserve signal reception in high-mobility or environments with imprecise CSI, facilitating graceful degradation. In contrast, narrow beams, with high gain, focus energy on the target, swiftly restoring transmission rates in scenarios involving low mobility or precise CSI. The integration of narrow and wide beams enables the system to adapt to varying conditions, mitigating performance decline under pressure and enhancing recovery speed, thus improving system resilience.

Typically, the dual-scale beam design incorporates two elements: channel parameter estimation and beam parameter design. It begins with extracting critical channel parameters from real-time channel state information (CSI). Unlike conventional system design, which straightforwardly employs CSI to formulate the problem, adjustment of MA positions also influences the wireless channel. Consequently, based on the CSI, the base station must estimate channel parameters, such as angle of arrival, delay, and gains of propagation paths, to develop a physics-based MA channel model. This model should quantify the link between antenna positions and corresponding wireless channels. According to the established MA channel model, the task of optimizing MA positions alongside beamforming is formulated across various channel conditions.

Wide Beam For Graceful Degradation: In instances where CSI is imprecise, particularly with fast-moving users, it becomes challenging to precisely align narrow beams to user locations. Such beam misalignment can lead to significant performance declines and inter-user interference. Consequently, strategies using wide beams and robust precoding are adopted to optimize average rates considering channel uncertainty. This approach combines wide beams for continuous coverage of mobile users

with interference-sensitive precoding to alleviate co-channel interference, thus reducing degradation.

Narrow Beam For Rapid Recovery: With accurate CSI, such as for low-mobility users, MA elements are strategically positioned to locations with improved channel conditions. Narrow beams then concentrate energy on target users, enhancing signal reception and reducing interference. This spatial and signal optimization focuses transmission energy into an accurate high-gain beam, improving the signal-to-interference-plus-noise ratio (SINR) or transmission rate. Thus, narrow beams facilitate high-quality communication.

Furthermore, deep neural networks (DNNs) can be applied to effectively optimize dynamic antenna setups in MA frameworks by modeling the nonlinear relationship between user locations and ideal antenna spacing for targeted beam properties, reducing complex real-time calculations. For particular user positions, the DNN directly predicts the best antenna spacing, greatly simplifying design and reducing computational demands.

Fig. 3 illustrates the system sum rate of narrow versus wide beams in high-mobility scenarios. High-speed mobility causes CSI to become outdated, impairing narrow beam performance when CSI is inaccurate. Consequently, narrow beams alone are insufficient for reliable communication. However, wide beams and robust precoding can notably enhance transmission rates and offer graceful performance degradation relative to perfect CSI. Therefore, MA-enabled dual-scale beams improve wireless communication efficiency and resilience across various conditions.

IV. DUAL-SCALE BEAM ADVANCED SENSING CAPABILITIES

This section discusses traditional sensing and its challenges within an integrated sensing and communication (ISAC) system. Subsequently, we explore the dual-scale sensing beam design for enhanced capabilities.

A. SENSING CHALLENGES

The ISAC system is a novel framework that integrates waveform design to simultaneously transmit sensing and communication signals within one frequency band, facilitating efficient spectrum resource management and collaborative use [13], [14]. Typically, ISAC utilizes MIMO technology to improve spectrum efficiency via parallel data streams. The antenna arrangement heavily influences both angular resolution and beamforming gain. While large antenna arrays offer high angular resolution, they also increase hardware costs and power consumption. Sparse arrays with large antenna spacing address this issue. Despite providing narrow beams, sparse arrays fall short in adapting to varying sensing requirements. Narrow, sharply focused beams enhance directional accuracy and spatial resolution, aiding in precise object detection and localization. Nonetheless, narrow beam sensing faces specific challenges. Without prior knowledge of object positions, locating targets with narrow beams becomes time-consuming and inefficient. Additionally, narrow beams experience spatial ambiguity in multi-target environments due to periodic phase differences. This periodic nature can cause different-angle targets to appear with identical or aliased phases, confusing multiple objects with a single target or obscuring true positions. These limitations complicate the maintenance of reliable sensing performance.

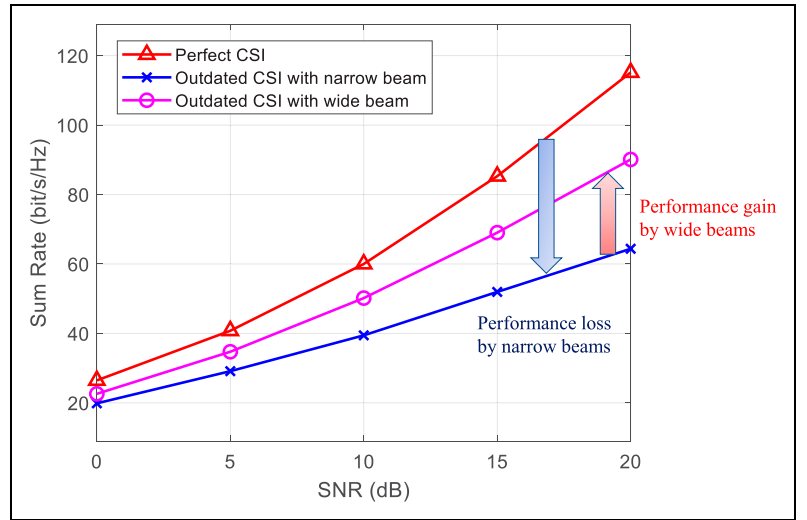


FIG. 3. Comparison of narrow and wide beams for high-mobility environment.

The MA-enhanced dual-scale beams introduce enhanced spatial adaptability. Narrow beams enable high-precision detection and localization within confined regions, whereas wide beams support rapid environmental scanning and broad area monitoring. By adjusting MA positions and precoding dynamically, dual-scale beams provide adaptable sensing—from detailed area inspection to wide-ranging situational awareness.

B. DUAL-SCALE SENSING BEAM DESIGN

The MAs enable the system to efficiently execute a hierarchical beam process for target acquisition, resembling a “search-and-focus” method. Initially, MAs are configured into wide beams, allowing rapid scanning over a large area with broad angular coverage. This phase detects the presence and general location of potential targets (single or multiple) within the area. Subsequently, the system shifts to high-resolution sensing by generating highly directional narrow beams. This adaptive dual-scale beams significantly improve sensing agility, accuracy, and efficiency in varying and changing environments. The detailed sensing process contains the following two phases.

Phase 1: Wide Beam Scanning: MA-enabled wide-beam sensing concentrates on developing efficient strategies for fast, large-scale environmental scanning and preliminary target identification. The main goal is to utilize the spatial adaptability of MAs to dynamically configure antenna elements in a compact manner. This setup extends angular coverage, enabling large-area scanning in one sweep. An appropriate beamwidth should be designed to acquire the initial sensing information of the number and rough positions of targets.

Phase 2: Narrow Beam Localization: Utilizing the wide beam scanning, the sensing system adjusts the MA positions to produce narrow beams for precise sensing. For a single target, MAs direct a high-gain narrow beam towards its approximate location, enhancing the precision of arrival angles. When encountering target clusters, the system tactically modifies beamwidth, employing beams narrow enough to achieve high angular resolution and spatial target separation, while wide enough to encompass the target cluster, thus reducing cross-interference and measurement

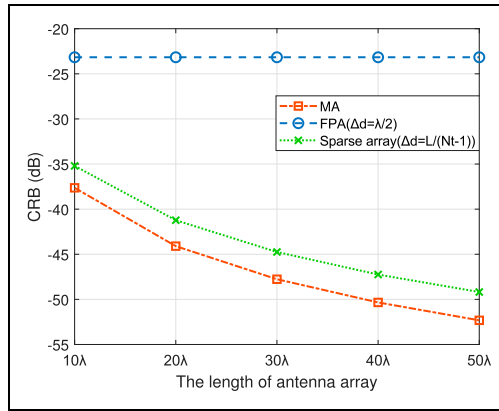


FIG. 4. CRB comparison of different antenna arrays.

ambiguity. This adaptive beamwidth control improves spatial separation, allowing for precise tracking of multiple objects simultaneously.

Both wide beam scanning and narrow beam localization require careful design of MA positions and precoding. A fundamental challenge is balancing beam coverage with sensing resolution. Wide beams offer broad coverage and low resolution, whereas narrow beams provide limited coverage but high resolution. In Phase 1, the objective of MA positions and precoding design is to minimize the expected Cramer-Rao bound (CRB) or the expected mean square error for angle estimation, with the expected CRB reflecting average outcomes within the scanning area and facilitating appropriate beam for specific regions. In Phase 2, given predetermined angles, MA positions and precoding are optimized to directly minimize the CRB of these angles, thus enhancing sensing accuracy. The essence lies in configuring MA elements based on real-time target data, such as the number of detected objects and their spatial distribution within the area. The flexibility of dual-scale beams through MA repositioning enhances sensing precision in varied and dynamic environments. The design of dual-scale beams for communication and sensing has been explored individually. In the ISAC system, when both communication and sensing signals are transmitted at once, their combined beam design exhibits similarities.

Moreover, AI technologies, including DNNs, can aid in designing dual-scale beam sensing during the two phases mentioned. In Phase 1, DNNs process wide-beam scan data to estimate target counts, approximate angles, and potential target clusters, then provide MA repositioning or precoding configurations for Phase 2. This approach decreases trial scans and alleviates spatial aliasing issues that narrow beams might encounter. During Phase 2, hybrids of model-based techniques and learning approaches can enhance angle/delay estimation and adapt across different environments, thereby facilitating real-time sensing in dynamic contexts.

Fig. 4 presents a comparison of the CRB for various antenna arrays. The results reveal that MAs consistently reach the minimum CRB, ensuring superior sensing accuracy. Conversely, the fixed half-wavelength array exhibits a notably high CRB, leading to significant performance degradation. Although the sparse antenna array approaches the MA's performance, its CRB remains comparatively high.

V. DUAL-SCALE BEAM STRENGTHENED SECURITY

In this section, we briefly introduce the physical layer security and its challenges. Then, we will discuss the security strengthened by the MA-enabled dual-scale beams.

A. PHYSICAL LAYER SECURITY

Wireless communication applications have proliferated, bringing significant data security risks due to the openness of wireless transmission. Traditionally, high-layer encryption ensures data security, with a central challenge being secret key distribution among numerous users. Physical layer security (PLS), harnessing intrinsic channel characteristics, has been widely researched as a viable solution for wireless security [15]. Typically, PLS encompasses keyless secure transmission and physical layer key generation. Keyless secure transmission, grounded in information-theoretic security, leverages the quality gap between legitimate and eavesdropping channels to ensure secure transmission through superposition coding. Physical layer key generation enables two legitimate users to generate symmetric keys from wireless channel fading, which relies on the reciprocity of legitimate channels and the disparity with eavesdropper channels. In both security types, spatial resources are crucial, thus multiple antennas are employed to generate narrow beams to increase channel quality gap or extend channel disparity.

However, extending PLS techniques to large-scale user environments presents substantial challenges. As the number of users grows, essential problems arise: superior channel condition construction and privacy key generation become more complex due to inter-user interference, the eavesdropping risks significantly increase, and channel reciprocity declines considerably. These challenges significantly compromise the effectiveness and security of current PLS approaches. Therefore, developing efficient, scalable, and low-complexity PLS solutions for massive connectivity is a critical research challenge. Wide beams can separate massive users into distinct regions, thereby reducing eavesdropping and interference. Within these regions, narrow beams can further ensure secure transmission. Combining narrow and wide beams mitigates their limitations, thus strengthening the security of wireless communications.

B. DUAL-SCALE SECURE BEAM DESIGN

Dual-scale beams present a promising solution to the challenges posed by a large number of users. Unlike their application in communications and sensing, where different scale beams are used for different scenarios, dual-scale beams are applied simultaneously in physical layer security transmission. Wide beams separate users into distinct regions, eliminating cross-region signal interference and reducing eavesdropping risks. After isolation, narrow beams are utilized for transmitting secure data or generating secret keys to/with fewer users within each region. Next, we will specifically explore wide beam isolation for user partitioning and narrow beam encryption to improve secure communication.

Wide Beam Isolation: The presence of massive users presents challenges in both keyless secure transmission and physical layer key generation. In keyless secure transmission, massive users bring substantial interference, with the risk of some users intercepting private signals. The transmission's security rate

is determined as the smallest rate differential between one user and others, potentially yielding a zero rate for some users. By devising broad beams to divide users into separate spatial zones, those in different beams are spatially segregated, diminishing interference and eavesdropping threats. In physical layer key generation, estimating reciprocal CSI necessitates orthogonal pilot signals, increasing pilot overhead beyond the channel's inherent duration and disrupting channel reciprocity. Narrow beams enable pilot signal reuse among users, diminishing pilot overhead, but reuse-induced estimation errors negatively impact the secret key rate. To mitigate interference, wide beams can group users, permitting pilot signal reuse among different groups while using orthogonal pilot signals to distinguish closely located users within each group.

Narrow Beam Encryption: Based on the wide beam spatial separation, narrow beams are further designed to enhance secure transmission. In keyless secure transmission, narrow beams help maximize the disparity between the rates of the legitimate user and the eavesdropper, thereby improving the secure transmission rate. When precise information about the eavesdropper's channel is lacking, robust designs for the worst-case scenario are considered to optimize the minimum rate across various potential eavesdropper locations. In physical layer key generation, the secret-key rate is determined by the mutual information between the legitimate users' channel observations, given the eavesdropper's observations. Narrow beams are designed to increase channel randomness and reciprocity between the legitimate pair, while diminishing correlation with potential eavesdroppers.

Both wide and narrow beams depend on the design of MA positions and effective precoding. Therefore, jointly optimizing MA positions and precoding for wide and narrow beams is crucial. Through iterative design, the system ensures a synergistic interaction between the beam scales. The complexity inherent in iterative design benefits from utilizing DNNs for predicting secrecy rates and mapping observed CSI/statistics to approximate optimal MA placements and precoders. DNNs are capable of identifying suspicious measurement patterns and forecasting probable eavesdropper locations. Based on these locations, DNNs can estimate secure transmission rates or secret-key rates considering potential eavesdroppers. Finally, DNNs model the transformation from observed CSI to MA positions and precoders that enhance secrecy metrics, refining beam design robustness and decreasing online optimization latency.

Fig. 5 illustrates the secret key generation rates across various schemes. With perfect CSI, the system attains the maximum secret key rate. Yet, with a high user count, orthogonal pilot signals induce delays and hinder channel reciprocity, while narrow beams are prone to interference. Consequently, these two methods witness a notable drop in secret key rates. Conversely, dual-scale beams significantly enhance the secret key rate, nearing perfect CSI performance.

VI. CONCLUSION AND FUTURE WORK

This article introduces MA-enabled dual-scale beams aimed at improving communication resilience, sensing, and security. By adjusting element positions, these MAs facilitate wide and narrow beams to

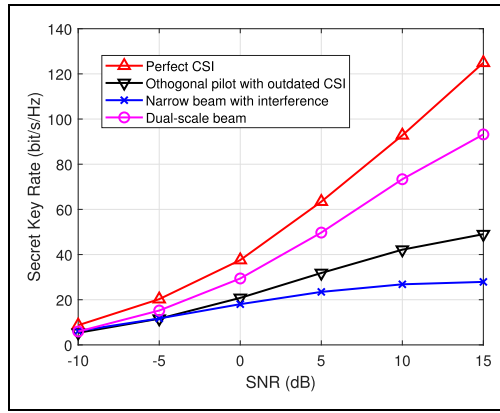


FIG. 5. Comparison of secret key generation for different schemes.

achieve effective beam control, thereby exploiting spatial degrees of freedom. We address challenges in communication, sensing, and security, and explore potential solutions through dual-scale beams. In wireless communication and sensing, the adaptability of dual-scale beams increases communication rates and sensing accuracy in varied and dynamic environments. For physical layer security, wide and narrow beams are combined to strengthen security. Further research is essential to address these challenges and advance the use of dual-scale beams for various environments.

A. FUTURE RESEARCH DIRECTIONS

Advanced Scenario Recognition: The dual-scale beams adapt to varying scenarios by adjusting accordingly. Therefore, recognizing scenarios is crucial. By utilizing techniques like DNN, scenario recognition should efficiently classify communication environments. This involves processing multi-type inputs, potentially including coarse user location distributions, historical interference patterns, or even simple sensor inputs. The accurate scenario recognition will allow the system further to balance between wide and narrow beams, maximizing the effectiveness and efficiency of the dual-scale beams in various environments.

Efficient Channel Information Acquisition: Unlike FAs, MA systems adjust antenna positions, leading to alterations in the corresponding CSI. Different antenna configurations result in different CSI, even with the same users. Therefore, inferring CSI for varying antenna setups from channel estimation is another challenge. To address this issue, a precise MA channel model is essential, as it plays a crucial role in system performance. The base station extracts channel parameters through estimation to construct CSI following the channel model. Given the complex relationship between CSI and antenna positions, artificial intelligence methods can be utilized.

Efficient MA Position Optimization: The complexity of optimizing MA positions and precoding in massive systems presents significant challenges. Future research will aim to develop low-complexity algorithms and frameworks that achieve near-optimal solutions. Promising strategies include: decoupling MA positioning and precoding tasks, integrating analytical models with neural networks for efficient optimization, and pre-calculating MA configuration patterns for typical scenarios, enabling quick online selection.

Robust Design: The design of MAs and precoding relies on CSI, and conversely, affects it. Hence, acquiring precise CSI at the BS is challenging. Given the uncertainty of CSI, a robust approach to designing MAs and precoding is feasible. In security contexts, when eavesdropper information is not available to the BS, the design can focus on optimizing the worst-case secure transmission rate to enhance system robustness.

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